Ex.No.1 **FORMATION OF BUS ADMTTANCE AND BUS IMPEDANCE MATRIX**
Date:

AIM

To develop a program to obtain Y_{bus} matrix for the given networks by the method of inspection.

SOFTWARE REQUIRED

MATLAB/ETAP/Power world simulator

PREREQUISITE QUESTIONS

1. Define sparsity in a matrix.
2. Recall the concept of singular matrix.
3. State the application of cofactor matrix.
4. Differentiate primitive admittance and bus admittance.
5. Define line charging admittance in a power system network

ALGORITHM

Step 1: Initialize [Y-Bus] matrix that is replace all entries by zero.

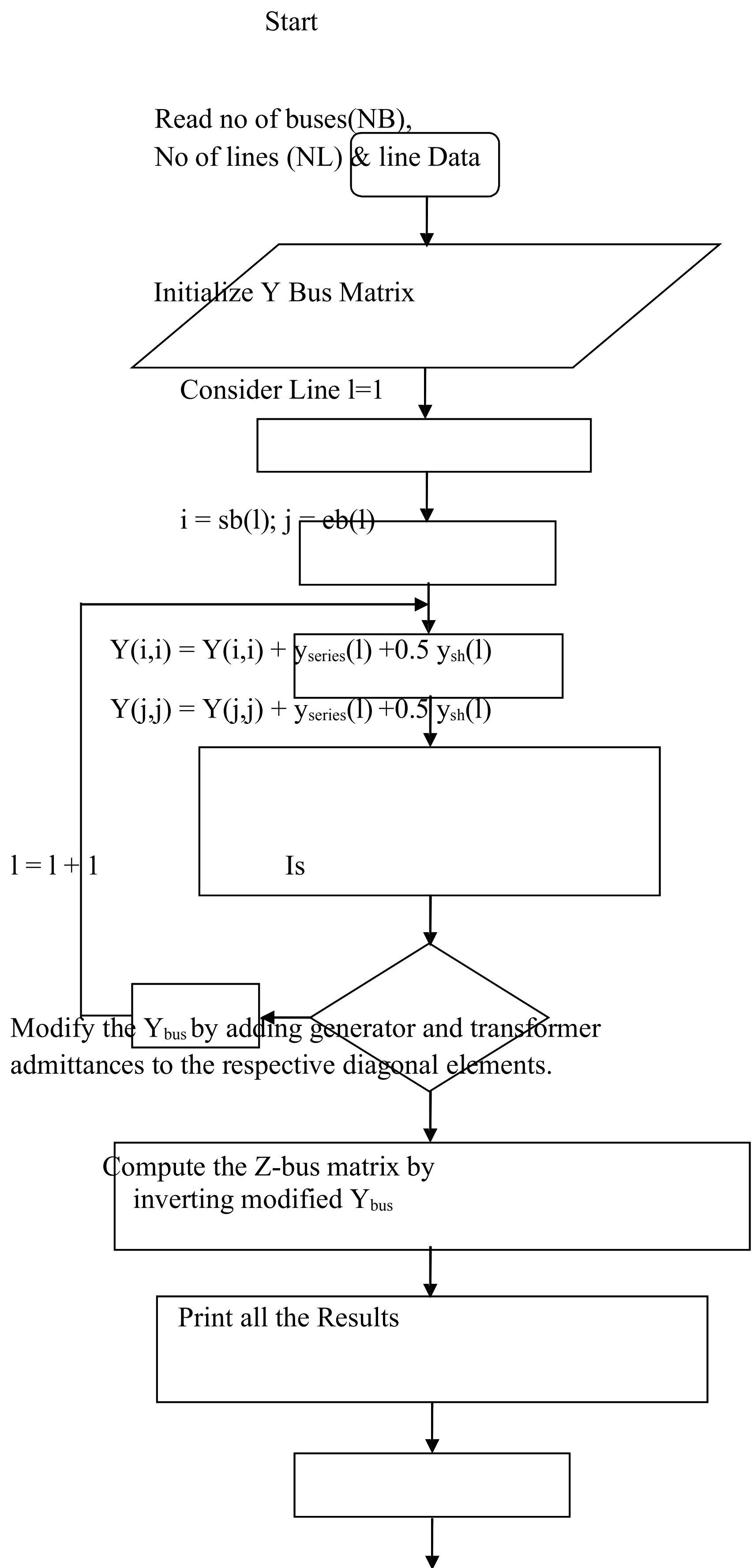
$Y_{ij} = Y_{ij} - y_{ij} = Y_{ji}$ = off diagonal element

Step 2: Compute $Y_{ii} = \sum_{j=1} y_{ij}$ = diagonal element.

Step 3: Modify the Y_{bus} matrix by adding the transformer and the generator admittances to the respective diagonal elements of Y bus matrix

Step 4: Compute the Z-Bus matrix by inverting the modified Y_{bus} matrix

FLOW CHART


Start
Read no of buses(NB),
No of lines (NL) & line Data
Initialize Y Bus Matrix
Consider Line l=1
i = sb(l); j = eb(l)
$Y(i,i) = Y(i,i) + y_{series}(l) + 0.5\ y_{sh}(l)$
$Y(j,j) = Y(j,j) + y_{series}(l) + 0.5\ y_{sh}(l)$
l = l + 1
Is
Modify the Y_{bus} by adding generator and transformer admittances to the respective diagonal elements.
Compute the Z-bus matrix by inverting modified Y_{bus}
Print all the Results

PROBLEM STATEMENT

The [Y-Bus] matrix is formed by inspection method for a four bus system. The line data and is given below.

LINE DATA

Line Number	SB	EB	Series Impedance (p.u)	Line charging Admittance (p.u)
1	1	2	0.10 + j0.40	j0.015
2	2	3	0.15 + j0.60	j0.020
3	3	4	0.18 + j0.55	j0.018
4	4	1	0.10 + j0.35	j0.012
5	4	2	0.25 + j0.20	j0.030

FORMATION OF Y-BUS BY THE METHOD OF INSPECTION

PROGRAM:

```
clc;
clear all;
n=input('Enter number of buses');
l=input('Number of lines');
s=input('1.Impedance or 2:Admittance');
ybus=zeros(n,n);
lc=zeros(n,n);
for i=1:l
a=input('Starting bus:');
b=input('Ending bus:');
t=input('Admittance or Impedance of line:');
lca=input('Line charging admittance:');
if(s==1)
y(a,b)=1/t;
else
y(a,b)=t;
end
y(b,a)=y(a,b);
lc(a,b)=lca;
lc(b,a)=lc(a,b);
end
for i=1:n
```

```
for j=1:n
if i==j
for k=1:n
ybus(i,j)=ybus(i,j)+y(i,k)+lc(i,k)/2;
end else
ybus(i,j)=-y(i,j);
end
end ybus
```

FORMATION OF Z-BUS BY THE METHOD OF INSPECTION

PROGRAM:

```
clc;
clear all;
n=input('Enter number of buses'); l=input('Number of
lines'); s=input('1.Impedance or 2:Admittance');
for i=1:l
a=input('Starting bus:');
b=input('Ending bus:');
t=input('Admittance or Impedance of line:');
lca=input('Line charging admittance:');
if(s==1)
y(a,b)=1/t
else
y(a,b)=t;
end
y(b,a)=y(a,b);
lc(a,b)=lca;
lc(b,a)=lc(a,b);
end
ybus=zeros(n,n);
for i=1:n
for j=1:n
if i==j
for k=1:n
ybus(i,j)=ybus(i,j)+y(i,k)+lc(i,k)/2;
end else
ybus(i,j)=-y(i,j);
end
ybus(j,i)=ybus(i,j);
end
end
zbus=(ybus)^(-1);
zbus
```

OUTPUT:

VIVA QUESTIONS:

1. Differentiate primitive admittance and bus admittance in network
2. Differentiate line charging and half line charging admittance based on Ferranti effect
3. List the two applications of Y bus
4. State the applications of Y bus.
5. Indicate the difference between driving point and transfer admittance in Y bus.

STIMULATING QUESTIONS:

1. State the reason for using Z bus in short circuit studies.
2. Why Y bus is sparse and Z bus is not sparse?

RESULT

Ex.No-2 **FORMATION OF BUS INCIDENCE AND LOOP INCIDENCE MATRIX**
Date:

AIM

To write a suitable program to form a bus incidence and loop incidence matrix for a given power system.

SOFTWARE REQUIRED

MATLAB/ETAP/Power world simulator

PREREQUISITE QUESTIONS

1. State the impedance of single line diagram.
2. Differentiate Bus and Node based on network behaviour.
3. Infer the importance of network topology determination.
4. Indicate the order of bus admittance matrix for a given network with ‘n’ nodes.

ALGORITHM

Step 1: Start the program

Step 2: Using the switch cases ask either bus incidence or loop incidence matrix to be found. If loop incidence matrix is to be formed go to step6.

Step 3: Ask number of system.

Step 4: To form bus incidence matrix, rules are
+1= If current is leaving the node
-1= If current is entering the node.
0= if branch is not incident to the node.

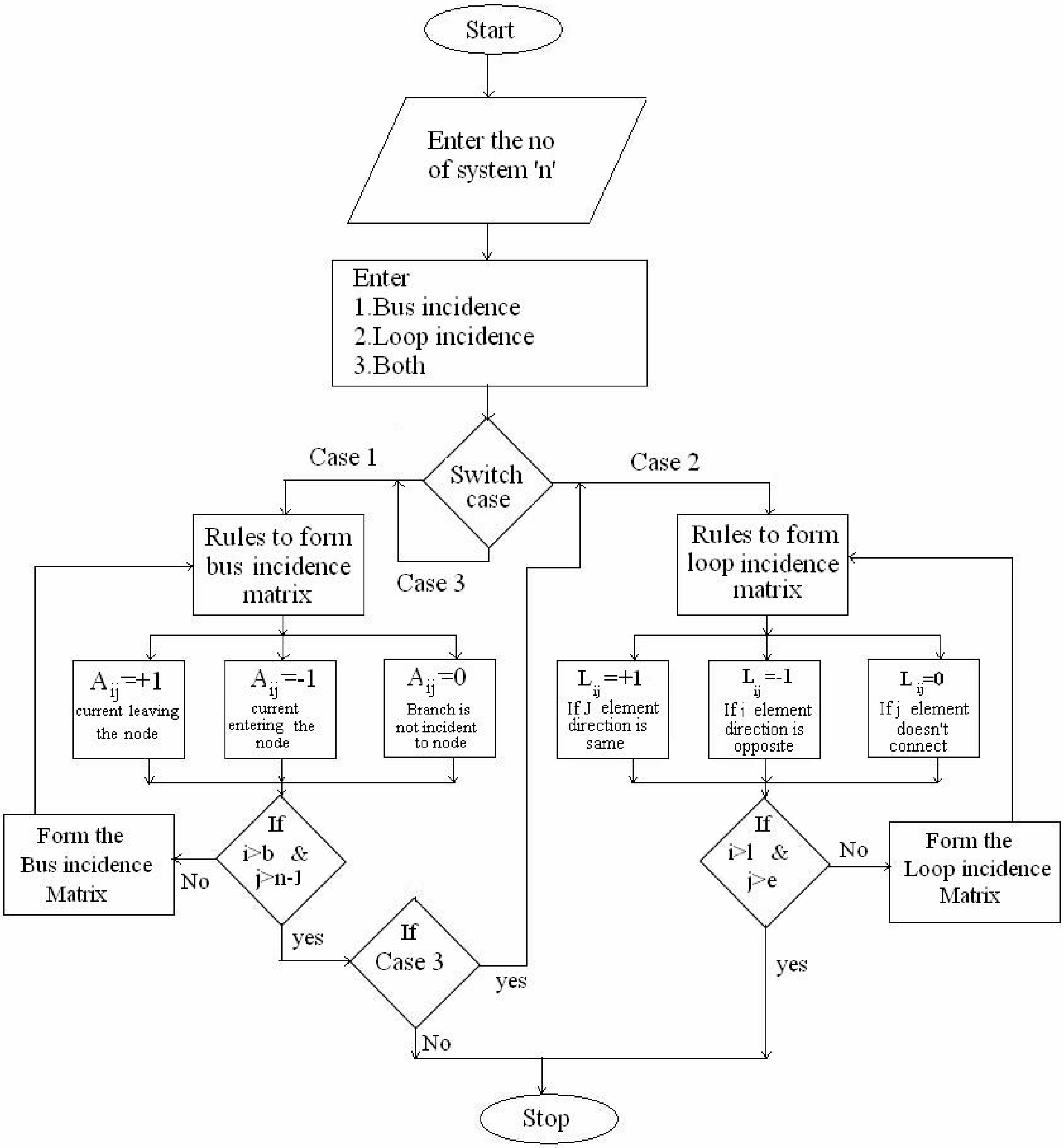
Step 5: Using this rule, form bus incidence matrix.

Step 6: To form loop incidence matrix, form loop of a graph drawn, size of the matrix and element node.

Step 7: The direction of the loop is direction of link, the matrix is corresponding element
+1=If the element direction is same as that of its corresponding loop.
-1=If the element is opposite direction.
0=If the element direction does not correspond to that loop.
Step 8: Using the above rules, form the loop incidence matrix and display it.

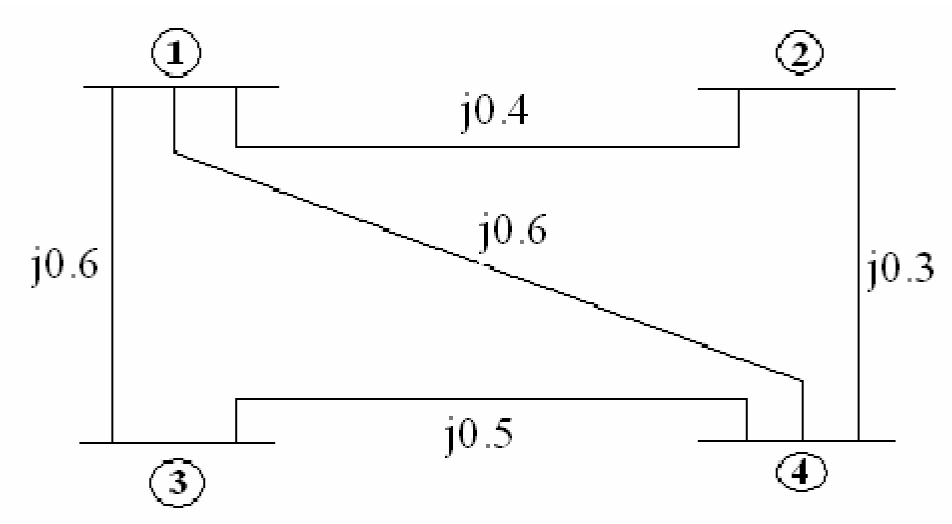
Step 9: Stop the program.

FLOW CHART



PROBLEM STATEMENT

Find the bus incidence matrix [A] and loop incidence matrix [L] for the four bus system shown in figure. Take ground as reference.



BUS INCIDENCE MATRIX PROGRAM:

```
clear;
clc;
n=input(' Enter the no of nodes:')
e=input('Enter the no of elements:')
for i=1:e
for j=1:n
fprintf(' \nClick the incidence of element %d with node %d',i,j);
reply=menu('choice','incident Away','incident towards','not incident');
if(reply==1)
a(i,j)=1;
end
if (reply==2)
a(i,j)=-1;
end
if(reply==3)
a(i,j)=0;
end
end
end
ref=input('\nEnter the reference node no:');
fprintf('\n The bus incidence matrix is =\n');
for i=1:e
for j=1:n

if(j~=ref)
fprintf('%d \t',a(i,j));
end
end
fprintf('\n');
```

```
end
```

LOOP INCIDENCE MATRIX PROGRAM:

```
clear;
clc;
y=input('Enter no of elements:');
l=input('Enter no of loops:');
for i=1:y
for j=1:l
fprintf('Is the element %d incident with loop %d?',i,j);
r=input('Enter 1 for YES and 0 for NO:');
if(r==1)
fprintf('Is the element %d directed with loop %d?',i,j);
s=input('Enter 1 for YES and 0 for NO:');
if(r==1)
c(i,j)=1;
end
if(s==0)
c(i,j)=-1;
end
end
if(r==0)
c(i,j)=0;
end
end
end
fprintf('The loop incidence matrix is:');
disp(c);
```

OUTPUT:

BUS INCIDENCE MATRIX

LOOP INCIDENCE MATRIX

VIVA QUESTIONS:

1. Differentiate Branch and Link in a network
2. Define a tree of the network
3. Indicate the difference between the matrices A and $\hat{A}$
4. Differentiate connected graph and oriented graph of a network
5. State the method for assigning orientation to a network elements

STIMULATING QUESTIONS:

1. Find the number of loops of a network with ‘n’ nodes and ‘m’ elements.
2. State the reason for omitting ground node while forming matrix A.

RESULT

Ex.No-3 **FORMATION OF BRANCH PATH MATRIX AND CUTSET MATRIX**
Date:

AIM

To write a program to obtain the branch path and cutest matrix using c++ program.

SOFTWARE REQUIRED

MATLAB/ETAP/Power world simulator

PREREQUISITE QUESTIONS

1. Indicate the number of branches in a network with ‘n’ nodes.
2. Infer the significance of primitive admittance of network elements.
3. Differentiate independence and mutually coupled elements.
4. List the merits of network matrices

ALGORITHM

Step 1: Start the program.

Step 2: Declare the variable and get the number of nodes and branches.

Step 3: Get the node number and also check whether that node is connected to that branch.

Step 4: If it is connected, then check the direction of element, if it is towards the node put ‘-1’ in branch path matrix else ‘1’.

Step 5**:** If it is not connected put ‘0’ in the matrix.

Step 6: Print the branch path matrix.

Step 7: Calculate the number of branches in the matrix.

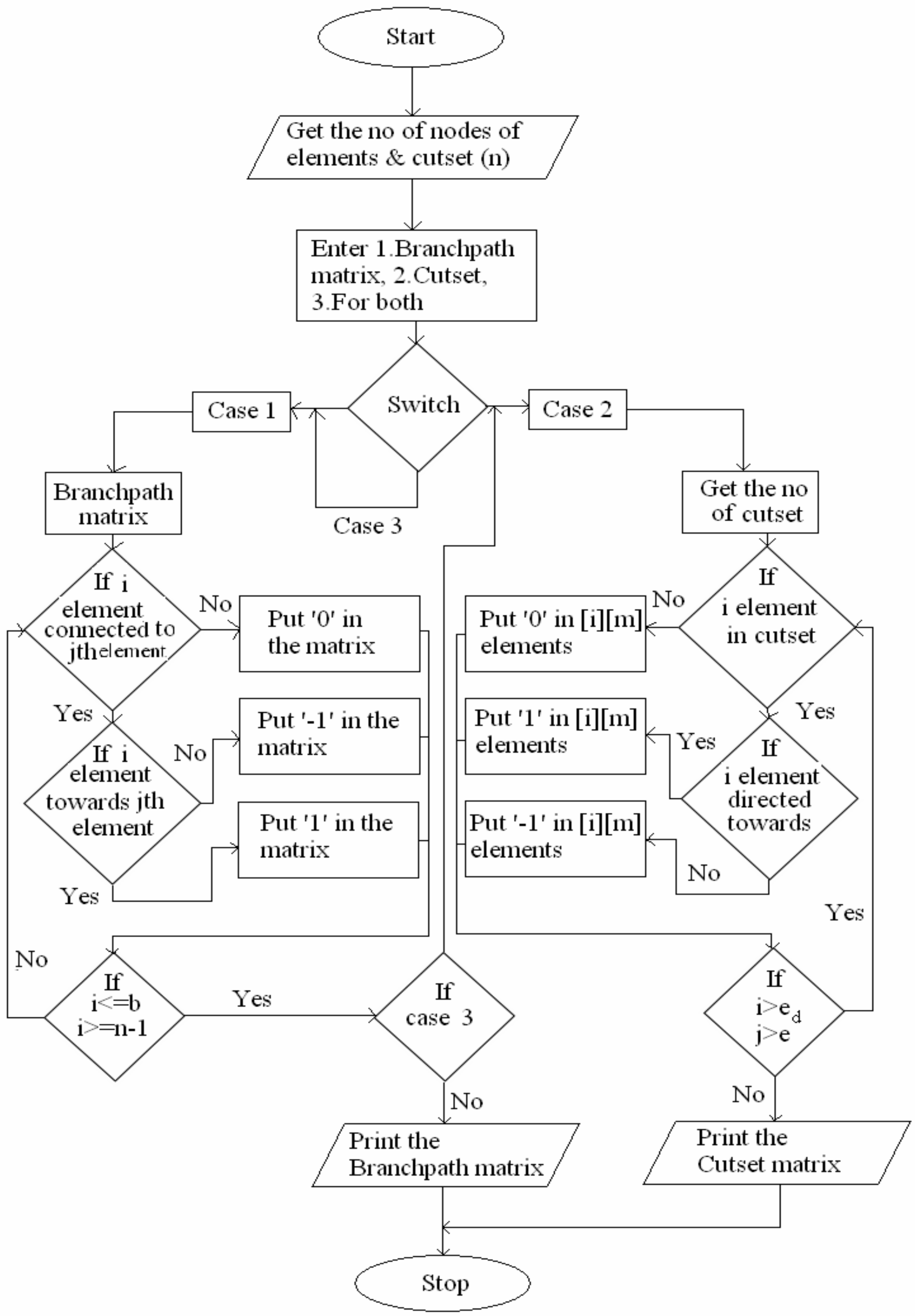
Step 8: Get the elements number & cutest number to check whether that element is present in the cutest or not.

Step 9: If the element is present check its direction & put ‘1’, if it is opposite direction with the branch direction else put ‘1’.

Step 10: Print the cutest matrix.

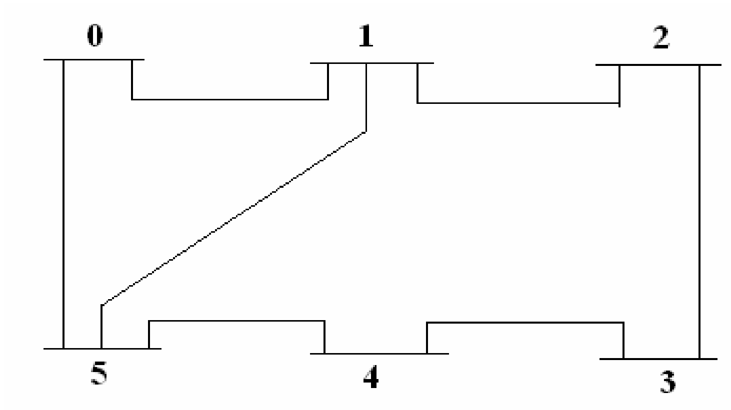
Step 11: Stop the program.

FLOW CHART



PROBLEM STATEMENT

Find the branch path matrix and cutset matrix of the given power system.



BRANCH PATH INCIDENCE MATRIX PROGRAM:

```
clear;
clc;
n=input('nodes');
b=input('branches');
for i=1:b
for j=1:n
fprintf('is the node %d connected to the branch %d',j,i);
p=input('enter 1 for yes and 0 for no');
if(p==1)
fprintf('is the direction of branch %d towards the node %d?:',i,j);
q=input('enter 1 for yes and 0 for no');
if(q==1)
k(i,j)=1;
end
if(q==0)
k(i,j)= -1;
end
end
if(p==0)
k(i,j)=0;
end
end
end
fprintf('the branch path incidence matrix is:');
for i=1:b
for j=1:n
fprintf('%d\t',k(i,j));
end
fprintf('\n');
end;
```

CUTSET MATRIX PROGRAM:

```
clear;
clc;
e=input('no of elements');
c=input('no of cutsets');
```

```
for i=1:e
for j=1:c
fprintf('Is element %d present in cutset %d',i,j);
p=input('enter 1 for yes and 0 for no');
if(p==1)
fprintf('Is the element %d directed along the direction of cutset%d?',i,j);
q=input('enter 1 for yes and 0 for no');
if(q==1)
a(i,j)=1;
end
if(q==0)
a(i,j)=-1;
end
end
if(p==0)
a(i,j)=0;
end
end
end
fprintf('the cutset matrix is:');
for i=1:e
for j=1:c
fprintf('%d\t',a(i,j));
end
fprintf('\n');
end
```

<u>OUTPUT:</u>

BRANCH PATH INCIDENCE MATRIX

CUT SET MATRIX

VIVA QUESTIONS :

1. Define a cutset.
2. Indicate the number of cutset in a given network with ‘n’ nodes.
3. Differentiate tree and co tree.
4. List two applications of cutset matrix.
5. List two applications of branch path matrix.

STIMULATING QUESTIONS:

1. Justify the usage of matrices in network topology estimator.
2. Identify the number of branches, number of links, number of cutset in a network with ‘n’ nodes and ‘m’ elements.

RESULT

Ex.No-4 **SOLUTION OF POWER FLOW AND RELATED PROBLEMS USING GAUSS SEIDEL METHOD**
Date:

AIM

To compute the program to find the load flow equations and solutions using Gauss Seidal method and Matlab.

SOFTWARE REQUIRED

MATLAB/ETAP/Power world simulator

PREREQUISITE QUESTIONS

1. List the types of buses in a power system network.
2. Differentiate generator bus and load bus.
3. Indicate the importance of slack bus in a network.
4. Justify the need for load flow studies.

ALGORITHM

Step 1**:** Assume the all buses voltages are 1+j0 except at slack bus. The voltage at Slack bus is Constant.

Step 2**:** Assume a suitable value for specified change is bus voltages which is used to compare the actual changes in bus voltage.

Step 3**:** Set the value of k=0.

Step 4**:** Set the bus count p=1.

Step 5: Check for slack bus. If it is slack bus go to step 12. Otherwise go to step 6.

Step 6**:** Check for generator bus. If so go to next step or go to step 9.

Step 7**:** Set $V_p^k = V_{p\,(Spec)}^k$ and phase V_p^k as the K^{th} iteration.

Calculate the reactive power of generator as following equations.

$$\mathbf{Q_{p.cal}^{k+1} = (-1)*Im\{(V_p^k)*[\sum_{q=1}^{p-1} Y_{pq}V_q^{k+1} + \sum_{q=p}^{n} Y_{pq}V_q^k]\}}$$

If $Q_{p.cal}^{k+1} < Q_{p.min}$ then $Q_p = Q_{p.min}$.
If $Q_{p.cal}^{k+1} > Q_{p.max}$ then $Qp = Q_{p.max}$.

Step 8**:** For Generator bus calculate the Bus voltage ($V_{p\,(temp)}^{k+1}$).

After calculating the Generator Bus Voltage go to step 11.

Step 9: For load bus calculate the value of V_p^{k+1} can be calculated.

Step 10: If acceleration factor is specified and modified the Bus voltage using the following equation.

$$\mathbf{V_{p\,(acc)}^{k+1} = V_p^k + \alpha(V_p^{k+1} - V_p^k).}$$

Then set $V_p^{k+1} = V_{p\,(acc)}^{k+1}$.

Step 11: Calculate the Change in bus voltage.

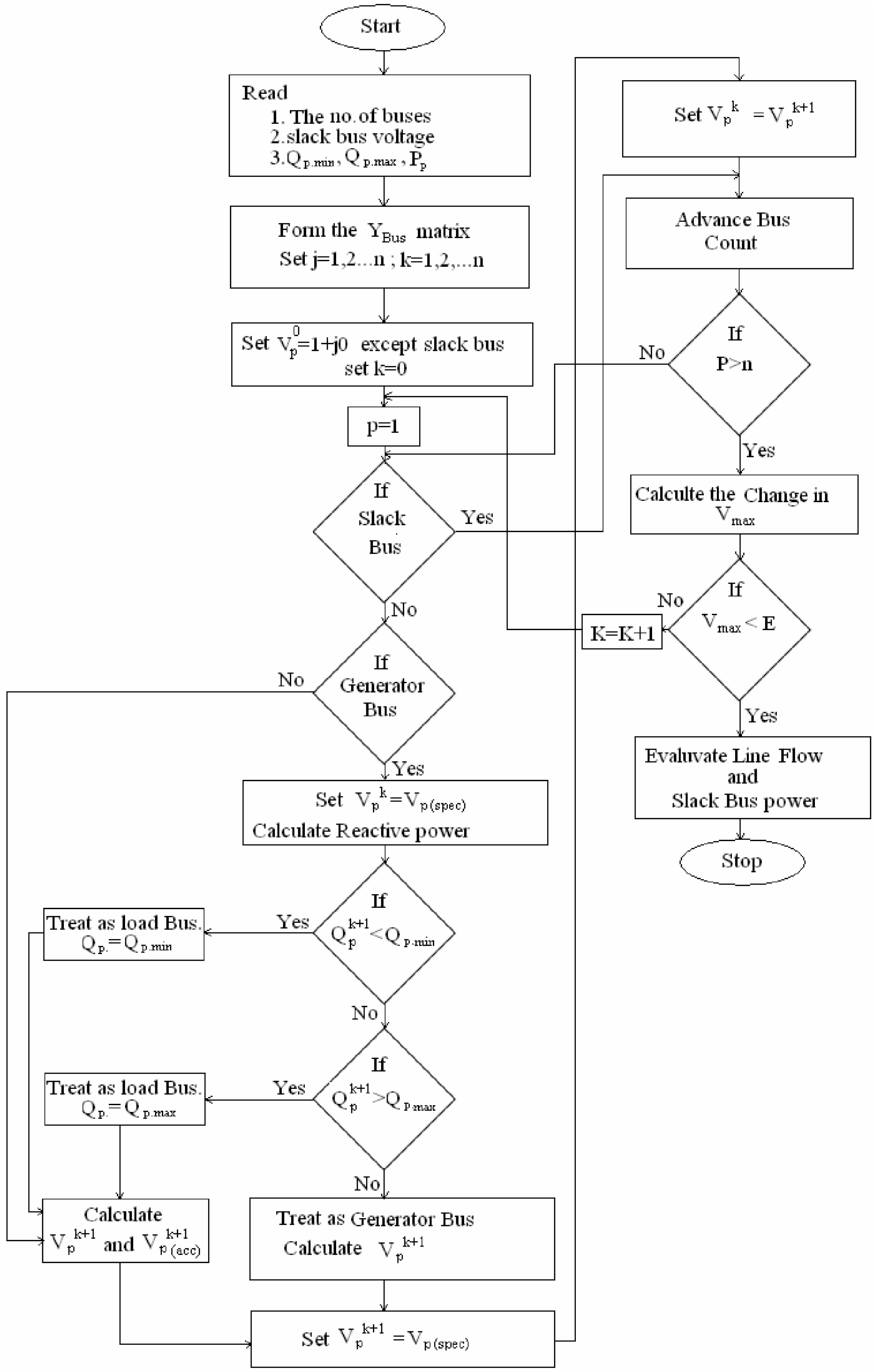
$$\mathbf{\Delta V_p^{k+1} = V_p^{k+1} - V_p^k.}$$

Step 12: Repeat 2 to 5 until all bus voltages are calculated.

Step 13: Find the largest among calculated voltage. If ΔV_{max} is less than prescribed value go to next step or increment iteration counts.

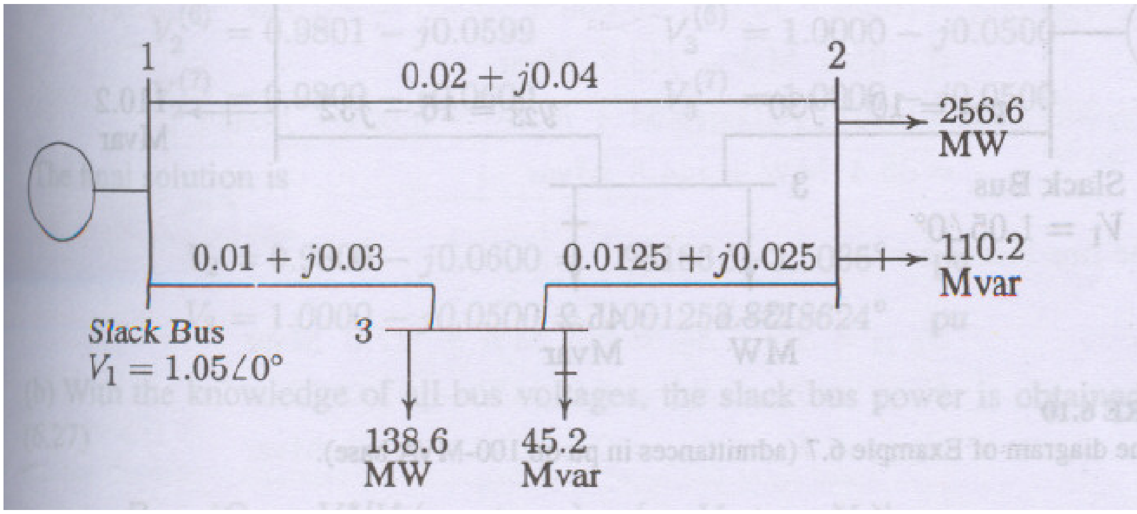
Step 14: Calculate line flows and slack bus power using bus voltages.

FLOW CHART



PROBLEM STATEMENT

Figure shows the one-line diagram of a simple three-bus power system with generation at bus 1. The magnitude of voltage at bus 1 is adjusted to 1.05 per unit. The scheduled loads at buses 2 and 3 are as marked on the diagram. Line impedances are marked in per unit on a 100-MVA base and the line charging susceptances are neglected.



<u>OUTPUT:</u>
Line Flows:

Line Losses:

VIVA QUESTIONS :

1. Define acceleration factor in Gauss Seidel method.
2. Indicate the voltage equation to be iterated in Gauss Seidel method.
3. Comment on the memory requirement of Gauss Seidal algorithm.

STIMULATING QUESTIONS:

1. Suggest a way to increase the convergence accuracy in Gauss Seidel method.
2. Indicate the drawbacks of solving load flow problem using Gauss Seidel method

RESULT

Ex.No-5 **SOLUTION OF POWER FLOW AND RELATED PROBLEMS USING**
Date: **NEWTON RAPHSON METHOD**

AIM

To compute the load flow equations for a multibus system using Newton Raphson method.

SOFTWARE REQUIRED

MATLAB/ETAP/Power world simulator

PREREQUISITE QUESTIONS

1. Differentiate DC and AC load flow methods.
2. Recall the advantages of using NR method in solving non linear problems.
3. Indicate the constraints to be satisfied while solving a network load flow.

ALGORITHM

Step 1: Assume the flat voltage for all buses except the slack bus and gen.bus.

Step 2: Set convergence criterion 'E' if the largest or the absolute of the residue is less than 'E' than The solution gets converged and the program gets terminated after calculating line flows, Otherwise repeat.

Step 3: Set the iteration count iter=1.

Step 4: Set the bus count n=1.

Step 5: Check if the bus is a slack bus if so goes to step 10.

Step 6: Calculate the real and reactive power Pp and Qp.

Step 7: Evaluate $\Delta P_p^k = P_{sp} - P_{pcal}$.

Step 8: Check if the bus is a generator bus, if so compare Q_p^k whether is there within the limits, violated and find the reactive power and treat bus as load bus if not calculate the voltage residue.

$$|\Delta V_p|^2 = |V_{p,}S_p|^2 - |V_{p,c}|^2$$

and go to step 10.

Step 9: Evaluate $\Delta Q_p^k = Q_{sp} - Q_{p,cal}$

Step 10: Advance bus count by 1 and check whether all the buses are counted if not go to step 8.

Step 11: Determine the largest absolute value or residue

Step 12: If largest of the residue is less than 'E' go to step 17.

Step 13: Find Jacobian matrix.

Step 14: Calculate new bus voltage

$e_p^{k+1}=e_p^k+\Delta e_p^k$

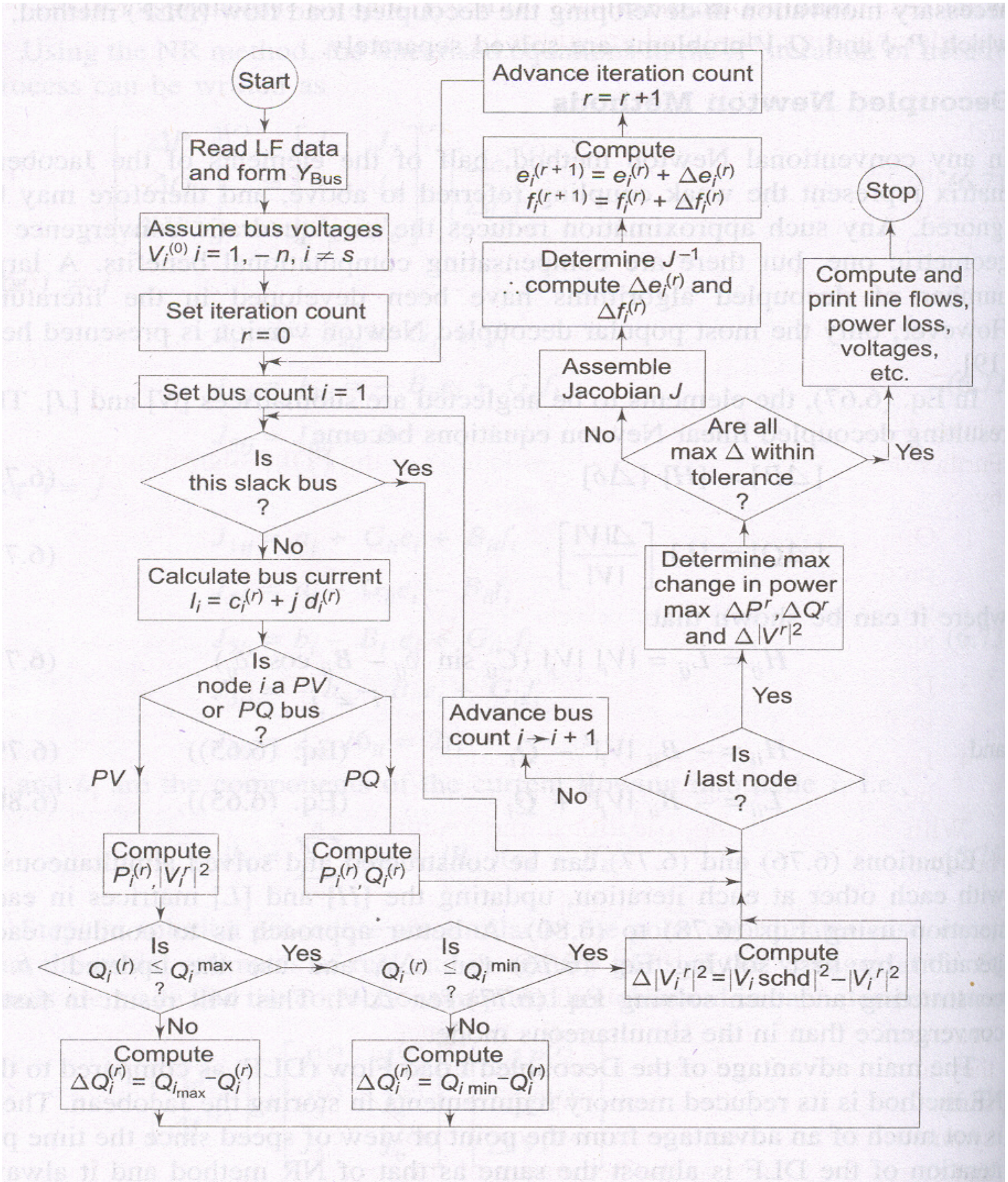
$f_p^{k+1}=f_p^k+\Delta f_p^k$

Step 15: Advance iteration count by 1& go to step 4.

Step 16: Evaluate bus & line voltage

Step 17: Print the result

FLOWCHART



PROBLEM STATEMENT

Bus data in p.u.

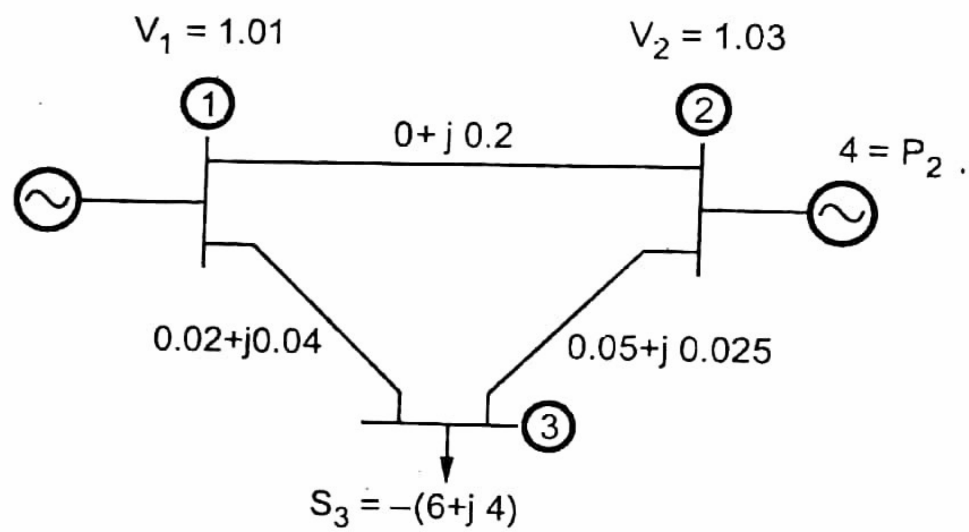
Bus Number	Type	Generator		Load		Voltage	Angle	Reactive Power Limit	
		P_G	Q_G	P_D	Q_D			Q_{min}	Q_{max}

1	Slack	0	0	0	0	1.01	0	0	0
2	PV	4.0	0	0	0	1.03	0	-2	2
3	PQ	0	0	6	4.0	1.0	0	0	0

Line data in p.u.

From	To	R p.u.	X p.u.	½ B1 p.u.	Tap Position
1	2	0	0.2	0	1
2	3	0.02	0.04	0	1
2	3	0.05	0.025	0	1

Base MVA: 100MVA



SOLUTION:

The line flows can be obtained as,

VIVA QUESTIONS:

1. Recall the elements of Jacobian 1 and Jacobian 3 in iterative equation.
2. Find the elements of Jacobian 4 for a network with one generator bus and three load buses apart from a slack bus.
3. Indicate the relationship between frequency and rotor displacement angle in an alternator.

STIMULATING QUESTIONS:

1. Compare the number of iterations required for convergence in Gauss Seidel and Newton Raphson method.
2. Compare the accuracy of Gauss Seidel and Newton Raphson method.

RESULT

Ex.No-6 **SOLUTION OF POWER FLOW AND RELATED PROBLEMS USING**
Date: **FAST DECOUPLED LOAD FLOW METHOD**

AIM

To compute the program to find the load flow equations and solutions using FDLF method.

SOFTWARE REQUIRED

MATLAB/ETAP/Power world simulator

PREREQUISITE QUESTIONS

1. Recall the control loops available in a synchronous generator.
2. State the relationship between reactive power delivered and terminal voltage of an alternator.
3. Differentiate generator bus and slack bus

ALGORITHM

Step 1: Read the system data.

Step 2: Initialize all the bus voltages.

Step 3: Form admittance matrix.

Step 4: Form B', B'' and then invert them.

Step 5: Compute $\Delta P = P^{spec} - P^{calc}$

Step 6: Check for convergence. If converged, set $P_{conv} = 1$, and go to step 10.

Step 7: Form Jacobian matrix H.

Step 8: If not converged, find $[\Delta\delta]$.

Step 9: Update δ value.

Step 10: Compute ΔQ.

Step 11: Check for convergence. If converged set $Q_{conv} = 1$, and go to step 15.

Step 12: Form [L].

Step 13: If not converged, find $[\Delta V]$.

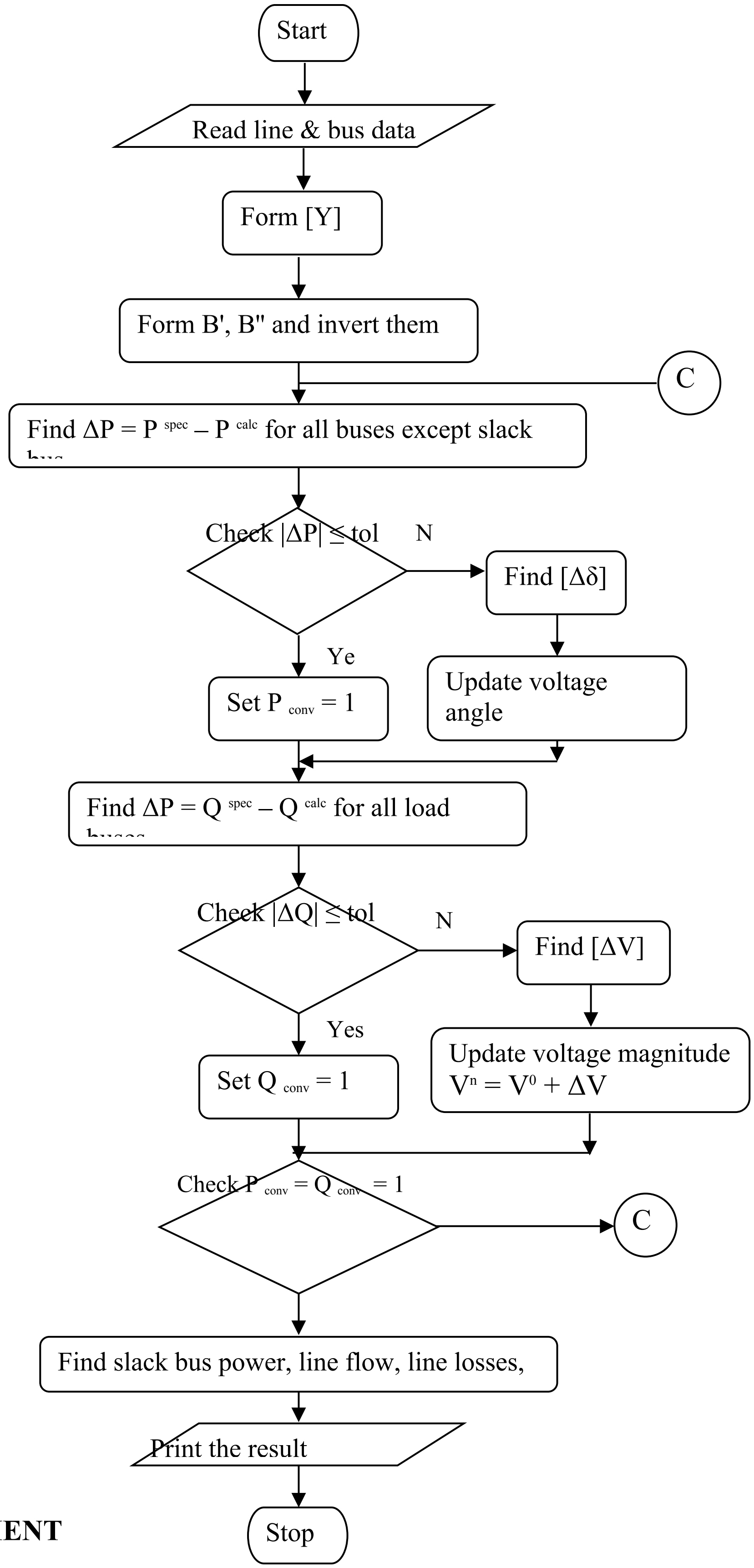
Step 14: Update $|V|$ value.

Step 15: If $P_{conv} = 1$ and $Q_{conv} = 1$, then solution is obtained and print the result.

Step 16: Else go to step 5.

Step 17: Stop.

FLOWCHART



PROBLEM STATEMENT

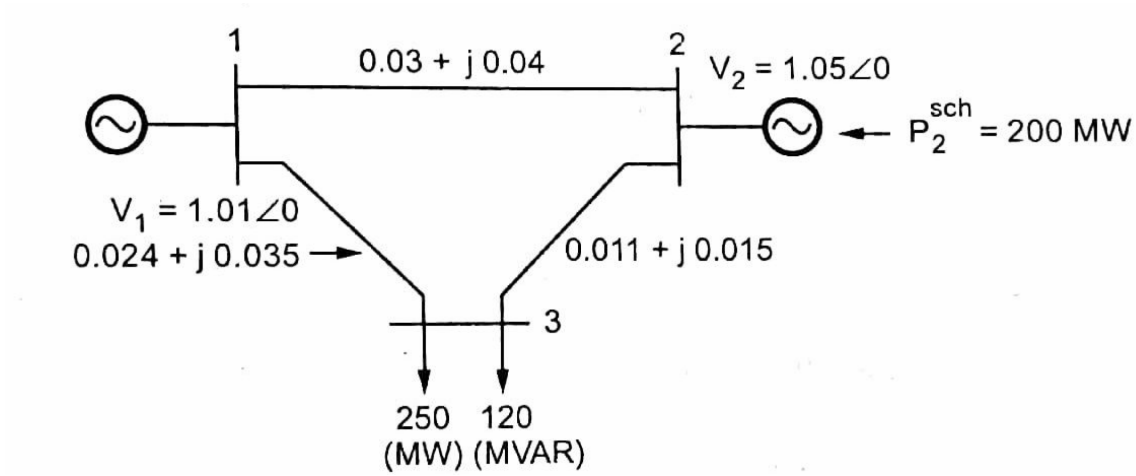
Bus data in p.u.

Bus Number	Type	Generator (MW) P_G	Q_G	Load (MW) P_D	Q_D	Voltage	Angle	Reactive Power Limit Q_{min}	Q_{max}
1	Slack	0	0	0	0	1.06	0	0	0
2	PQ	0	0	270	130	1.0	0	0	0
3	PV	230	0	0	0	1.04	0	0	230

Line data in p.u.

From	To	R p.u.	X p.u.	½ B1 p.u.	Tap Position
1	2	0.03	0.04	0	1
2	3	0.011	0.015	0	1
1	3	0.024	0.035	0	1

Base MVA: 100MVA



SOLUTION:

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VIVA QUESTIONS:

1. Indicate the Jacobian J2 and J3 of a network 1 slack bus and 1 generator bus and 2 load buses using FDLF method
2. Indicate the reason for Jacobian sparsity in FDLF method.

STIMULATING QUESTIONS

1. Establish the relationship between real power and voltage in an alternator during transient disturbance

RESULT

Ex.No-7 **SHORT CIRCUIT ANALYSIS**
Date:

AIM

To obtain the short circuit analysis for a IEEE 14 bus system.

SOFTWARE REQUIRED

MATLAB/ETAP/Power world simulator

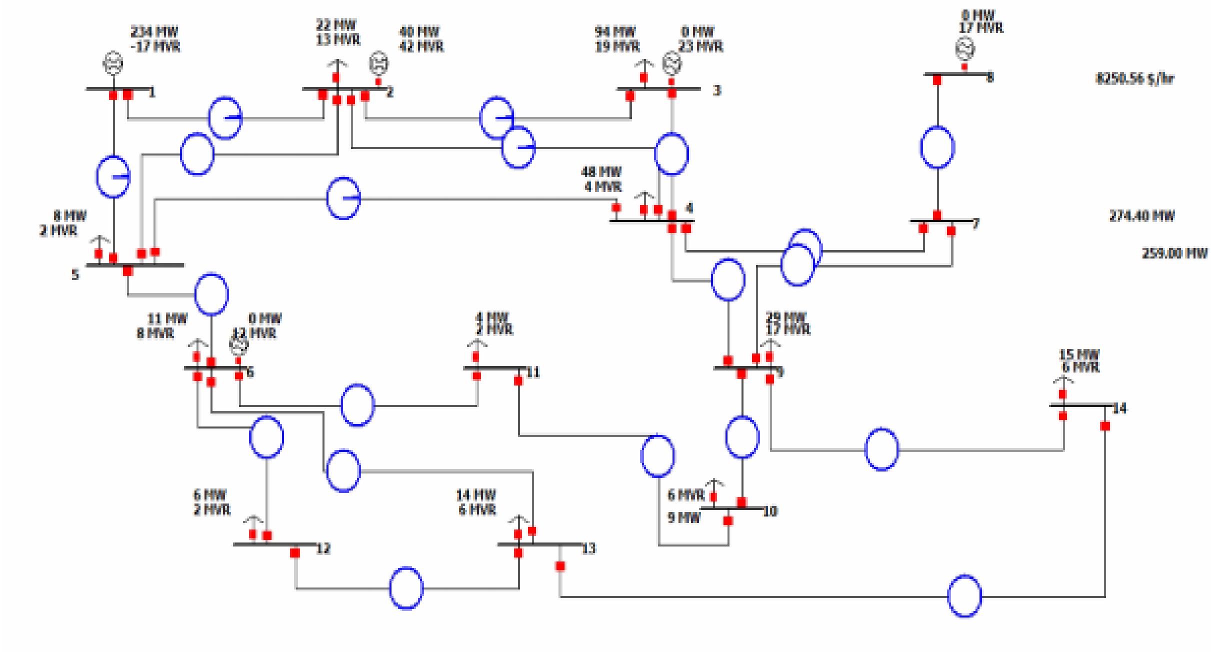
PREREQUISITE QUESTIONS

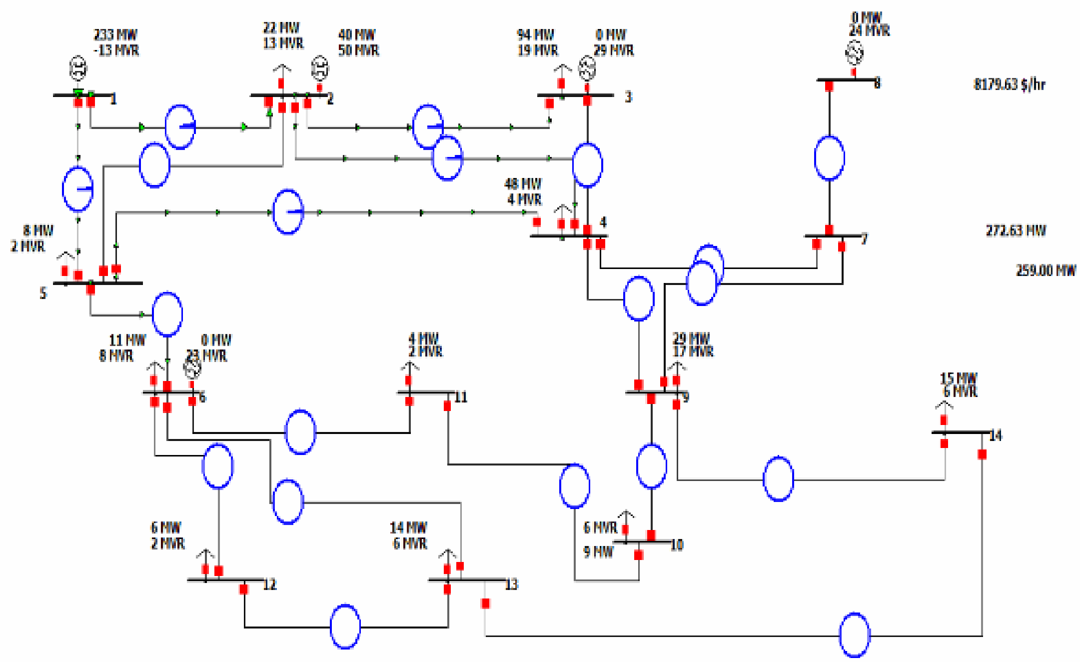
1. Classify the power system faults based on voltage symmetry
2. Indicate the magnitude of current flowing through the neutral conductor during three phase fault.
3. Differentiate symmetrical and unsymmetrical faults based on fault severity.

Procedure :

- Create a new file in edit mode by selecting File - New File.
- Browse the components and build the bus system.
- Execute the program in run mode by selecting tools-fault analysis.
- Select the fault on which bus and calculate.
- Tabulate the results.

14 bus system :





OUTPUT

VIVA QUESTIONS:

1. Indicate the equation to determine short circuit capacity at any point in a network.
2. Justify the need to determine fault current.
3. State the application of short circuit analysis.

STIMULATING QUESTIONS:

1. Differentiate line to line and line to ground fault based on probability of occurrence.
2. What will happen if a light bulb is connected between phase and earth.

RESULT:

Ex.No-8 **ECONOMIC DISPATCH IN POWER SYSTEMS**
Date:

AIM

To write a program to compute the economic dispatch control by iteration method using C program.

SOFTWARE REQUIRED

MATLAB/ETAP/Power world simulator

PREREQUISITE QUESTIONS

4. Differentiate Economic load dispatch and unit commitment.
5. List any 3 constraints in unit commitment problem.
6. Mention the stages of load dispatching in a power system

ALGORITHM

Step1: Start the program.

Step2: Set the initial value such that TFRs co-efficient should be less than the Chosen by us.

Step3: Calculate the power Pload for load=1, 2, 3….n
Where,
n- The number of plants connected to the load.

Step4: Calculate the tolerance value by using the formula.

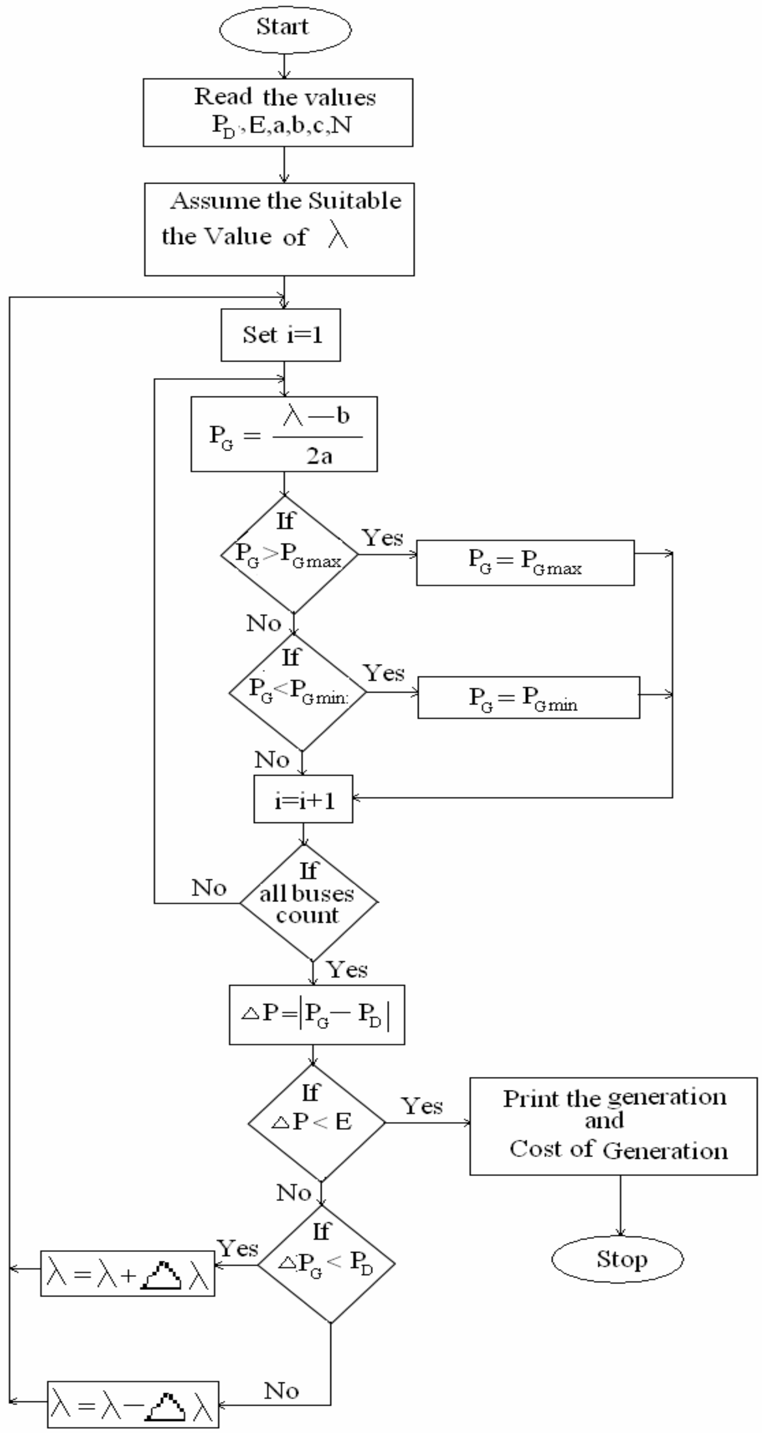
$$\varepsilon = P_r - \sum_{load=1}^{n} P_{load}$$

Step5: If the tolerance values negligible stop the program otherwise continue the Iteration.

Step6: Print the result.

Step7: Stop the program.

FLOWCHART



<u>OUTPUT</u>:

VIVA QUESTIONS:

1. List the constraints in Economic load dispatch.
2. State the Khun-Tucker condition for optimality.
3. Differentiate local minima and global minima in an Economic Load Dispatch problem
4. List the inequality constraints to be satisfied while solving load flow problem

STIMULATING QUESTIONS:

3. Examine the significance of Lambda in Economic load dispatch problem.
4. Determine the coordination equation for a system with power losses.

RESULT:

Ex.No-9 **TRANSIENT STABILITY ANALYSIS**
Date:

AIM

To analyze the modeling aspects of synchronous machine and Transient stability of Multimachine power systems.

SOFTWARE REQUIRED

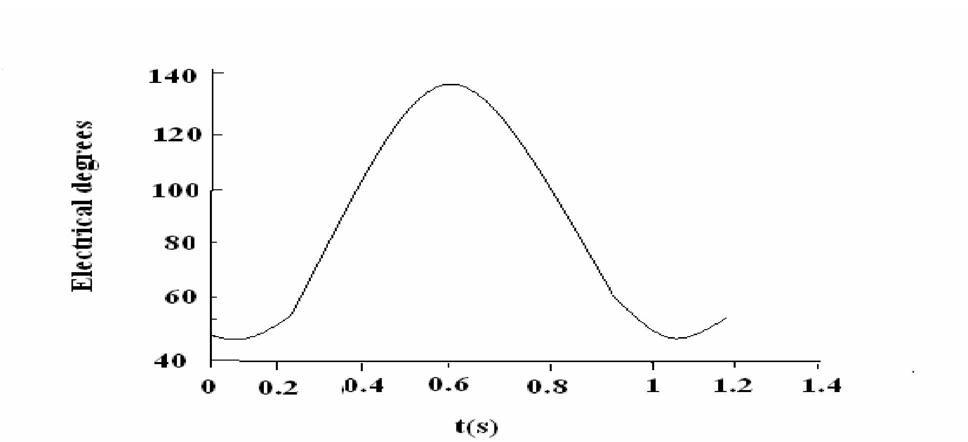
MATLAB/ETAP/Power world simulator

PREREQUISITE QUESTIONS

1. Define stability
2. Recall the equation to estimate power transferred between any two nodes in a networks.
3. Differentiate steady state and transient state stability.
4. State the characteristics of an infinite bus used for stability estimation.

ASSUMPTIONS

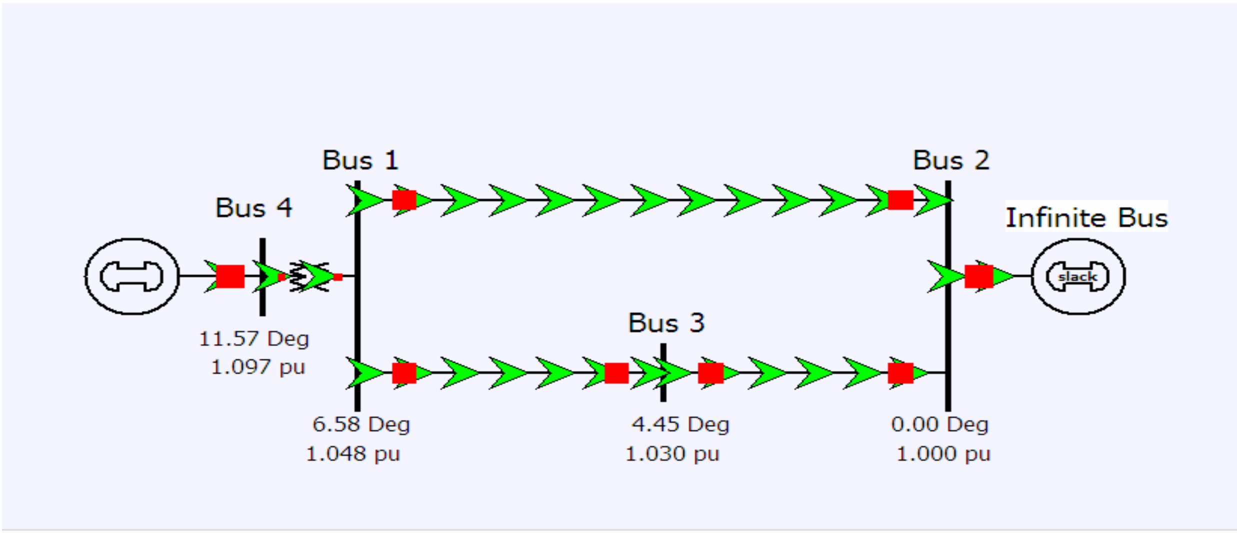
i. Generator saliency neglected is $X_d=X_q;X_d'=X_q'$
ii. All resistances are neglected.
iii.Damping is neglected
iv. Generators are represented by internal voltage behind the transient reactance.



Swing curve

MULTIMACHINE TRANSIENT STABILITY – SAMPLE PROBLEM

Consider a 60 HZ machine for which H=2.7MJ/MVA and it is initially operating in steady state with input and output of 1 p.u. and an angular displacement of 45 electrical degree with respect to an infinite bus bar. Upon occurance of a fault, assume that the input remains constant and the output is given by $P_e= \delta/90^o$. Calculate and plot the swing curve by the step-by-step method ii. Using the time interval Δt=0.05 s.up to t=1 s.step-by-step method –ll using the time interval Δt=0.05 s and upto t=1s.



<u>SOLUTION</u>

VIVA QUESTIONS:

1. Define critical clearing angle.
2. Interpret the stability of a system from a curve plotted between rotor displacement angle and time.
3. State the significance of power angle curve
4. Indicate the limits for δ in a stable system.

STIMULATING QUESTIONS:

1. What will happen if a generator with rated capacity 2000MW is subjected to a sudden load addition of 1500MW?

RESULT